



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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Experiment-2.2

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Semester: 5th

Subject Name: DAA

UID: 23BCS11196

Section/Group: 23KRG-2B

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Subject Code: 23CSH-301

1. **Aim:** Develop a program and analyze complexity to implement subset-sum problem using Dynamic Programming.
2. **Objective:** Objective is to implement subset-sum problem using Dynamic programming.
3. **Input/Apparatus Used:** Standard sum value is taken in order to find subsets of that sum using dynamic programming.

4. Procedure/Algorithm: Pseudocode:

So we will create a 2D array of size $(arr.size() + 1) * (target + 1)$ of type boolean.

The state $DP[i][j]$ will be true if there exists a subset of elements from $A[0...i]$ with sum value = „j“. The approach for the problem is:

if $(A[i-1] > j)$ $DP[i][j]$

= $DP[i-1][j]$

else

$DP[i][j] = DP[i-1][j] \text{ OR } DP[i-1][j-A[i-1]]$

1. This means that if current element has value greater than „current sum value“ we will copy the answer for previous cases
2. And if the current sum value is greater than the „ith“ element we will see if any of previous states have already experienced the sum=“j“ OR any previous states experienced a value „j – $A[i]$ “ which will solve our purpose.

5. Code:

```
Code.cpp ×
Code.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  bool subsetSum(int arr[], int n, int sum) {
5      bool dp[n+1][sum+1];
6      for(int i=0;i<=n;i++) dp[i][0]=true;
7      for(int j=1;j<=sum;j++) dp[0][j]=false;
8
9      for(int i=1;i<=n;i++){
10         for(int j=1;j<=sum;j++){
11             if(arr[i-1] > j)
12                 dp[i][j] = dp[i-1][j];
13             else
14                 dp[i][j] = dp[i-1][j] || dp[i-1][j-arr[i-1]];
15         }
16     }
17     return dp[n][sum];
18 }
19
20 int main() {
21     int n, sum;
22     cout << "Enter number of elements: ";
23     cin >> n;
24     int arr[n];
25     cout << "Enter elements: ";
26     for(int i=0;i<n;i++) cin >> arr[i];
27     cout << "Enter target sum: ";
28     cin >> sum;
29
30     if(subsetSum(arr, n, sum))
31         cout << "Subset with given sum exists";
32     else
33         cout << "Subset with given sum does not exist";
34 }
```



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6. Output:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\3rd Year\Semester-5\DAA_23BCS11196_KRG-2B\EXP-2.2> cd "d:\3rd Year\Semester-5\DAA_23BCS11196_KRG-2B\EXP-2.2\" ; if ($?) {
g++ Code.cpp -o Code } ; if ($?) { .\Code }
Enter number of elements: 4
Enter elements: 3
2
7
1
Enter target sum: 6
Subset with given sum exists
```